

Beyond X™

LV Soft Starter

LVSS 시리즈



LS ELECTRIC

Soft Starter

Soft Starter 사용 장점

- 기동 시 돌입전류(Inrush Current)를 최소화.
  - 모터 기동시 VOLTAGE DROP 최소화
  - 기동전류 최소화로 계통보호
  - 변압기 및 케이블 용량감소
  - 모터 보호 : 권선 히팅 최소화
- Soft Starting으로 기계장치 스트레스 최소화
  - 모터 기동시 모터와 기계장치에 인가되는 충격을 최소화
  - 장비 고장발생 횟수 저감 으로 유지보수 비용 절감
- 최적의 모터보호 기능
  - 운전상태 표시기능 : 각 상전류, 과부하율, Event Record등 Soft Starter 실시간 상태 확인
  - 단락, 과부하, 과전류, 결상, 모터 온도 보호 등 보호기능 내장.
- GREEN 제품(SCR-전력용 반도체) 적용
  - 구리, 알루미늄 등 적용 배제에 따른 탄소 배출 최소화
- 화재 예방 : Reactor 발화로 인한 화재.
  - 리액터 기동의 경우 높은 전류가 발생되고, 장시간 운전하게 되면 리액터의 발열 및 소손이 발생할 수 있음.
- Water Hammer 현상 제거 : Soft Stop 기능.
  * Water Hammer 현상 : 관로 안의 액체 운동상태를 급격히 변화시킴으로써 일어나는 압력파 현상.
- Harmonic 발생 최소화 : IEEE 519 만족.
  * IEEE 519 : 고조파 기준관리 (고조파 왜곡 규제)
  - 유도성 부하에서 발생하는 전압 왜곡파형을 끊어줌으로써 Harmonic 발생을 감소시키며, 인버터와는 달리 기동이 완료되면 Bypass mode로 전환되어서 고조파가 발생하지 않음.





Low Voltage Soft Starter

LVSS 시리즈







Contents

- Soft Starter의 장점 02
- LVSS 시리즈의 장점 04
- 선택 가이드 06
- Specifications 07
- 옵션 12
- 셋팅 방법 13
- Application 14
- 외형치수 15

LVSS Series

LVSS 시리즈의 장점



간편한 조작방법

• 시인성이 뛰어나고, 조작이 쉬운 Full Color Touch Screen(3.5인치)



4단계 파라메타

Application (적용 장비) 선택 → 모터 정격 전류 선택 → 제어 모드 선택 → 제어 전원 선택

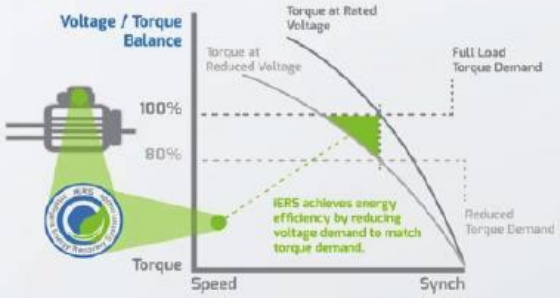
다국어 설정 가능

• Korean, Vietnam, Japanese, Russian, Turkish, Arabic, Chinese 등 17개 국어 표시 가능



에너지 절약

• iERS – Intelligent Energy Saving System (에너지 절약 시스템)



모터 경부하 시, 역률을 감지하여 출력전압을 제어함으로서 모터가 해당 부하에서 속도를 유지하는데 정확한 양의 에너지만 소비하도록 하는 에너지 절약시스템입니다. 압축기, 냉장고, 펌프잭, 성형기 및 냉각기와 같은 애플리케이션은 일반적으로 부하가 적은 경우 에너지 소비량의 최대 32%를 절약할 수 있습니다.

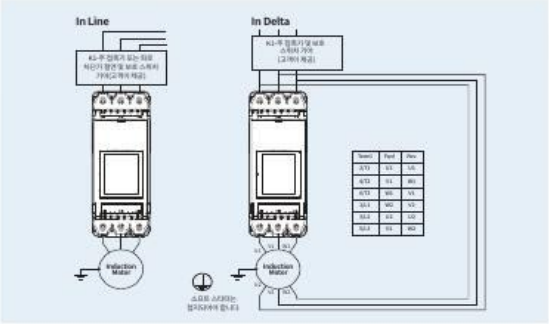
USB 사용 파라메타

• USB 사용 파라메타 Uploading / Downloading 가능



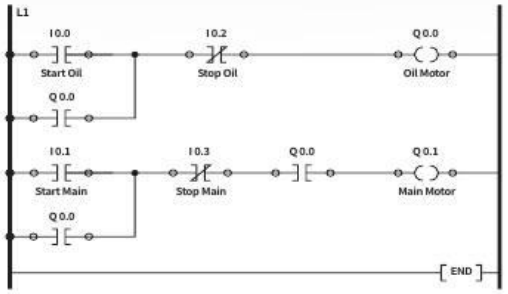
In Delta 결선 가능

• In Delta 결선 시, 한 단계 높은 모터용량 적용 가능



PLC Function

• Input, Output 단자를 programing 할 수 있으며, 신호 입력 및 출력에 대한 응답 시간 조절 가능



선택 가이드

LS-LVSS 제품은 200~480V에서 사용가능하고, 주파수는 50/60Hz 공용입니다.
이에 모터용량(kW) 기준으로 제품을 선택하시기를 바랍니다.

In Line				Select Model			Control Voltage
A	IEC (kW)			Trip Class 10	Trip Class 20	Trip Class 30	
	230 V	400 V	440 V	5 starts/hour @ 50°C			
17	4	7.5	7.5	LS-LVSS-101	VMX-SGY-103	LS-LVSS-105	LS-LVSS-101~305: DC24V AC110V AC220V
22	5.5	11	11	LS-LVSS-103	LS-LVSS-105	LS-LVSS-107	
29	7.5	15	15	LS-LVSS-105	LS-LVSS-107	LS-LVSS-109	
35	7.5	18.5	18.5	LS-LVSS-107	LS-LVSS-109	LS-LVSS-111	
41	11	22	22	LS-LVSS-109	LS-LVSS-111	LS-LVSS-113	
55	15	30	30	LS-LVSS-111	LS-LVSS-113	LS-LVSS-115	
66	18.5	37	37	LS-LVSS-113	LS-LVSS-115	LS-LVSS-117	
80	22	45	45	LS-LVSS-115	LS-LVSS-117	LS-LVSS-201	
100	30	55	55	LS-LVSS-117	LS-LVSS-201	LS-LVSS-203	
132	37	75	90	LS-LVSS-201	LS-LVSS-203	LS-LVSS-205	
160	45	90	110	LS-LVSS-203	LS-LVSS-205	⬇	
195	55	110	132	LS-LVSS-205	⬇	⬇	
3 stars/hour @ 50°C							
160	45	90	110	⬇	⬇	LS-LVSS-301	
195	55	110	132	⬇	LS-LVSS-301	LS-LVSS-303	
242	75	132	150	LS-LVSS-301	LS-LVSS-303	LS-LVSS-305	
302	90	160	185	LS-LVSS-303	LS-LVSS-305	LS-LVSS-307-1 LS-LVSS-307-2	
361	110	200	220	LS-LVSS-305	LS-LVSS-307-1 LS-LVSS-307-2	LS-LVSS-309-1 LS-LVSS-309-2	
430	132	250	280	LS-LVSS-307-1 LS-LVSS-307-2	LS-LVSS-309-1 LS-LVSS-309-2	⬇	
500	160	280	315	LS-LVSS-309-1 LS-LVSS-309-2	⬇	⬇	LS-LVSS-307~505-1: AC110V
3 stars/hour @ 40°C							
430	132	250	280	⬇	⬇	LS-LVSS-401-1 LS-LVSS-401-2	LS-LVSS-307~505-2: AC220V
500	160	280	315	⬇	LS-LVSS-401-1 LS-LVSS-401-2	LS-LVSS-403-1 LS-LVSS-403-2	
610	200	355	375	LS-LVSS-401-1 LS-LVSS-401-2	LS-LVSS-403-1 LS-LVSS-403-2	LS-LVSS-501-1 LS-LVSS-501-2	
722	220	400	425	LS-LVSS-403-1 LS-LVSS-403-2	LS-LVSS-501-1 LS-LVSS-501-2	LS-LVSS-503-1 LS-LVSS-503-2	
850	280	500	530	LS-LVSS-501-1 LS-LVSS-501-2	LS-LVSS-503-1 LS-LVSS-503-2	LS-LVSS-505-1 LS-LVSS-505-2	
960	315	560	600	LS-LVSS-503-1 LS-LVSS-503-2	LS-LVSS-505-1 LS-LVSS-505-2	-	
1080	355	630	670	LS-LVSS-505-1 LS-LVSS-505-2	-	-	

Specifications

LS-LVSS-101~117



Designations	3-phase SCR Energy Saving Motor Controller
Current	Size 1 up to 100A
Bypass	Internally Bypassed
Motor Protection	Full I^2T Motor Overload with memory Current limit set at 4.5x to meet IEC standards
Supply Voltage	208, 230, 400, 460 volts as standard Allowing for varying supply frequency 45-65 Hz
Standard Duty Rating	Trip Class 10 3xmotor current – 23sec, 3.5xmotor current – 17sec 5 starts/hour
Frame Sizes	Size 1
Enclosure Type	Up to 100A IP20/NEMA 1 - with finger guard
Terminal Position	Terminal position/spacing to match common contactors and circuit breaker
Terminal Construction	Cage Clamp
Control Voltage	24Vdc & 110/230Vac
Ambient Operating Conditions	-20°C to 50°C UP to 60°C with de rating
iERS Energy Saving Technology	YES
In Delta/6 wire connections	YES
Display Technology	3.5" Colour-TFT Touch Screen
Key pad	Optional fill color touch keypad available for remote/door fitting
Languages	English, Arabic, Chinese (Mandarin simplified), Dutch, French German, Greek, Italian, Japanese, Korean, Polish, Portuguese Russian, Serbian, Spanish, Turkish, Vietnamese
Input/Output	3x NO programmable output relays 1x NC programmable output relays 3x programmable digital inputs 1x PTC thermistor input 1x 0-10V/4-20mA Analogue input 1x 0-10V/4-20mA Analogue output 1x USB
Communications	Standard: Modbus RTU Optional: Anybus providing Ethernet IP, Modbus TCP & Profibus DP
Data Logging	Equipped with 4GB memory, allowing the logging of over 3.5 milli on events Date of first start (warranty date) Total number of soft starts Number of soft stops Number of error events Display event logs Export data in CSV format
EU/IEC Legislations	IEC 60947-4-2: 2012 Low Voltage Directive Battery Directive Energy Using Products/Energy Related Product Directives
Environmental	Products comply to REACH, SVHC. RoHS and WEEE
Standards	CE, CETL, KC
Field Service ability	Firmware upgrade from USB port

Specifications

LS-LVSS-201~205



Designations	3-phase SCR Energy Saving Motor Controller
Current	Size 2 up to 195A
Bypass	Internally Bypassed
Motor Protection	Full I^2T Motor Overload with memory Current limit set at 4.5x to meet IEC standard
Supply Voltage	208, 230, 400, 460 volts as standard Allowing for varying supply frequency 45-65 Hz
Standard Duty Rating	Trip Class 10 3 × 23 or 3.5 × 17 5 starts/hour
Frame Sizes	Size 2
Enclosure Type	Up to 195A IP20/NEMA 1 - with finger guard
Terminal Position	Terminal position/spacing to match common contactors and circuit breaker
Terminal Construction	Cage Clamp
Control Voltage	24Vdc & 110/230Vac
Ambient Operating Conditions	-20°C to 50°C UP to 60°C with de rating
iERS Energy Saving Technology	YES
In Delta/6 wire connections	YES
Display Technology	3.5" Colour-TFT Touch Screen
Key pad	Optional fill color touch keypad available for remote/door fitting
Languages	English, Arabic, Chinese (Mandarin simplified), Dutch, French German, Greek, Italian, Japanese, Korean, Polish, Portuguese Russian, Serbian, Spanish, Turkish, Vietnamese
Input/Output	3x NO programmable output relays 1x NC programmable output relays 3x programmable digital inputs 1x PTC thermistor input 1x 0-10V/4-20mA Analogue input 1x 0-10V/4-20mA Analogue output 1x USB
Communications	Standard: Modbus RTU Optional: Anybus providing Ethernet IP, Modbus TCP & Profibus DP
Data Logging	Equipped with 4GB memory, allowing the logging of over 3.5 milli on events Date of first start (warranty date) Total number of soft starts Number of soft stops Number of error events Display event logs Export data in CSV format
EU/IEC Legislations	IEC 60947-4-2: 2012 Low Voltage Directive Battery Directive Energy Using Products/Energy Related Product Directives
Environmental	Products comply to REACH, SVHC. RoHS and WEEE
Standards	CE, CETL, KC
Field Service ability	Firmware upgrade from USB port

LS-LVSS-301~309



Designations	3-phase SCR Energy Saving Motor Controller
Current	Size 3 up to 500A
Bypass	Internally Bypassed
Motor Protection	Full I^2T Motor Overload with memory Current limit set at 4.5x to meet IEC standard
Supply Voltage	208, 230, 400, 460 volts as standard Allowing for varying supply frequency 45-65 Hz
Standard Duty Rating	Trip Class 10 3 × 23 or 3.5 × 17 3 starts/hour
Frame Sizes	Size 3
Enclosure Type	Above 195A IP00/NEMA 1 Optional Finger Guards up to 195A to ensure fill IP20 enclosures
Terminal Position	Terminal position/spacing to match common contactors and circuit breaker
Terminal Construction	Plain Busbar
Control Voltage	24Vdc & 110/230Vac (LS-LVSS-301 to LS-LVSS-305) 110Vac or 230Vac (LS-LVSS-307 to LS-LVSS-309)
Ambient Operating Conditions	-20°C to 50°C UP to 60°C with de rating
iERS Energy Saving Technology	YES
In Delta/6 wire connections	YES
Display Technology	3.5" Colour-TFT Touch Screen
Key pad	Optional fill color touch keypad available for remote/door fitting
Languages	English, Arabic, Chinese (Mandarin simplified), Dutch, French German, Greek, Italian, Japanese, Korean, Polish, Portuguese Russian, Serbian, Spanish, Turkish, Vietnamese
Input/Output	3x NO programmable output relays 1x NC programmable output relays 3x programmable digital inputs 1x PTC thermistor input 1x 0-10V/4-20mA Analogue input 1x 0-10V/4-20mA Analogue output 1x USB
Communications	Standard: Modbus RTU Optional: Anybus providing Ethernet IP, Modbus TCP & Profibus DP
Data Logging	Equipped with 4GB memory, allowing the logging of over 3.5 milli on events Date of first start (warranty date) Total number of soft starts Number of soft stops Number of error events Display event logs Export data in CSV format
EU/IEC Legislations	IEC 60947-4-2: 2012 Low Voltage Directive Battery Directive Energy Using Products/Energy Related Product Directives
Environmental	Products comply to REACH, SVHC. RoHS and WEEE
Standards	CE, CETL, KC
Field Service ability	Firmware upgrade from USB port

Specifications

LS-LVSS-401~403



Designations	3-phase SCR Energy Saving Motor Controller
Current	Size 4 up to 722A
Bypass	No Bypass
Motor Protection	Full I^2T Motor Overload with memory Current limit set at 4.5x to meet IEC standard
Supply Voltage	208, 230, 400, 460 volts as standard Allowing for varying supply frequency 45-65 Hz
Standard Duty Rating	Trip Class 10 3 × 23 or 3.5 × 17 3 starts/hour
Frame Sizes	Size 4
Enclosure Type	Above 195A IP00/NEMA 1 Optional Finger Guards up to 195A to ensure fill IP20 enclosures
Terminal Position	Terminal position/spacing to match common contactors and circuit breaker
Terminal Construction	Plain Busbar
Control Voltage	110Vac or 230Vac
Ambient Operating Conditions	-20°C to 40°C UP to 60°C with de rating
iERS Energy Saving Technology	YES
In Delta/6 wire connections	YES
Display Technology	3.5" Colour-TFT Touch Screen
Key pad	Optional fill color touch keypad available for remote/door fitting
Languages	English, Arabic, Chinese (Mandarin simplified), Dutch, French German, Greek, Italian, Japanese, Korean, Polish, Portuguese Russian, Serbian, Spanish, Turkish, Vietnamese
Input/Output	3x NO programmable output relays 1x NC programmable output relays 3x programmable digital inputs 1x PTC thermistor input 1x 0-10V/4-20mA Analogue input 1x 0-10V/4-20mA Analogue output 1x USB
Communications	Standard: Modbus RTU Optional: Anybus providing Ethernet IP, Modbus TCP & Profibus DP
Data Logging	Equipped with 4GB memory, allowing the logging of over 3.5 milli on events Date of first start (warranty date) Total number of soft starts Number of soft stops Number of error events Display event logs Export data in CSV format
EU/IEC Legislations	IEC 60947-4-2: 2012 Low Voltage Directive Battery Directive Energy Using Products/Energy Related Product Directives
Environmental	Products comply to REACH, SVHC. RoHS and WEEE
Standards	CE, CETL, KC
Field Service ability	Firmware upgrade from USB port

LS-LVSS-501~505



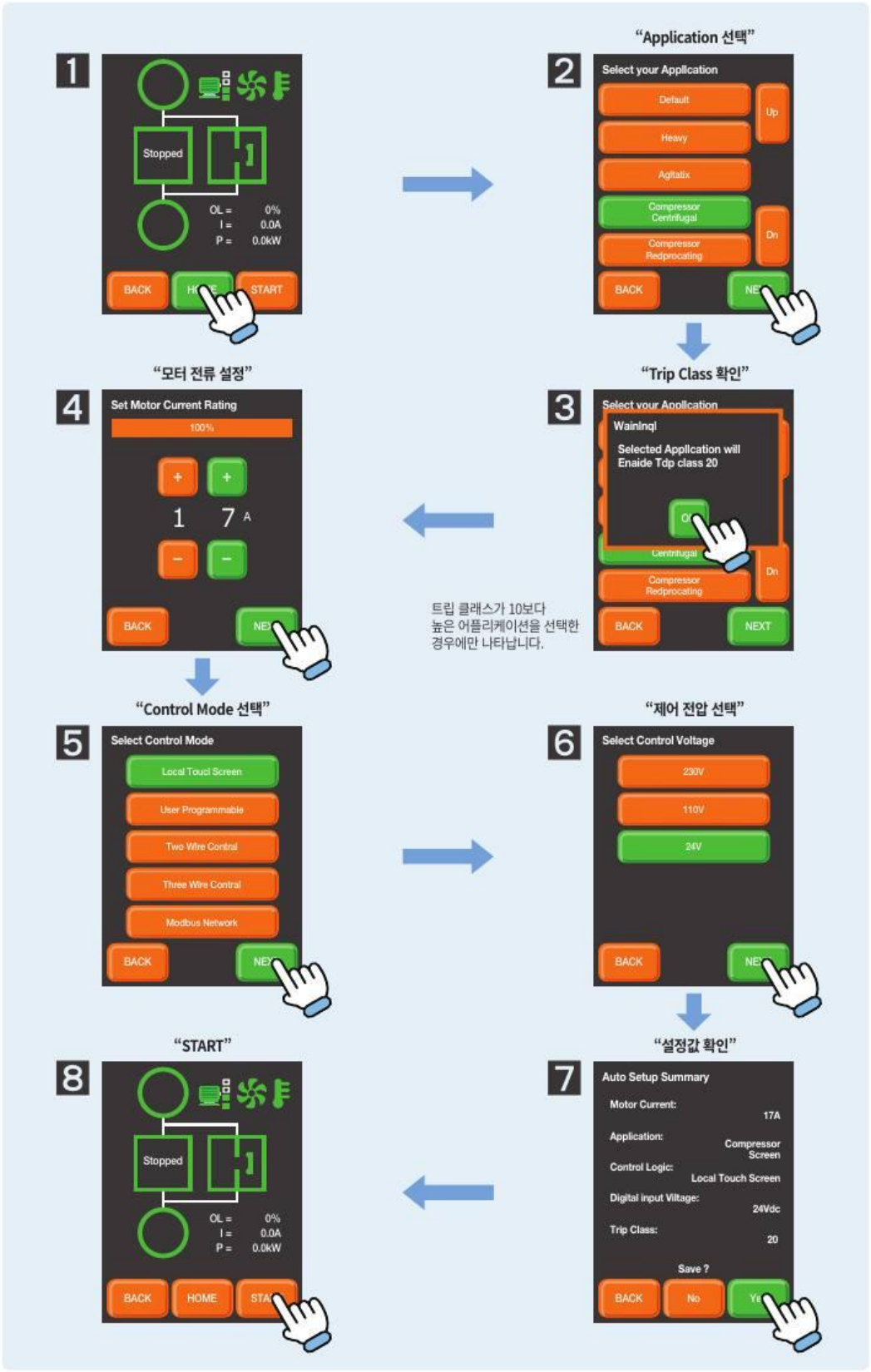
Designations	3-phase SCR Energy Saving Motor Controller
Current	Size 5 up to 1080A
Bypass	No Bypass
Motor Protection	Full I^2T Motor Overload with memory Current limit set at 4.5x to meet IEC standard
Supply Voltage	208, 230, 400, 460 volts as standard Allowing for varying supply frequency 45-65 Hz
Standard Duty Rating	Trip Class 10 3 × 23 or 3.5 × 17 3 starts/hour
Frame Sizes	Size 5
Enclosure Type	Above 195A IP00/NEMA 1 Optional Finger Guards up to 195A to ensure fill IP20 enclosures
Terminal Position	Terminal position/spacing to match common contactors and circuit breaker
Terminal Construction	Plain Busbar
Control Voltage	110Vac or 230Vac
Ambient Operating Conditions	-20°C to 40°C UP to 60°C with de rating
iERS Energy Saving Technology	YES
In Delta/6 wire connections	YES
Display Technology	3.5" Colour-TFT Touch Screen
Key pad	Optional fill color touch keypad available for remote/door fitting
Languages	English, Arabic, Chinese (Mandarin simplified), Dutch, French German, Greek, Italian, Japanese, Korean, Polish, Portuguese Russian, Serbian, Spanish, Turkish, Vietnamese
Input/Output	3x NO programmable output relays 1x NC programmable output relays 3x programmable digital inputs 1x PTC thermistor input 1x 0-10V/4-20mA Analogue input 1x 0-10V/4-20mA Analogue output 1x USB
Communications	Standard: Modbus RTU Optional: Anybus providing Ethernet IP, Modbus TCP & Profibus DP
Data Logging	Equipped with 4GB memory, allowing the logging of over 3.5 milli on events Date of first start (warranty date) Total number of soft starts Number of soft stops Number of error events Display event logs Export data in CSV format
EU/IEC Legislations	IEC 60947-4-2: 2012 Low Voltage Directive Battery Directive Energy Using Products/Energy Related Product Directives
Environmental	Products comply to REACH, SVHC. RoHS and WEEE
Standards	CE, CETL, KC
Field Service ability	Firmware upgrade from USB port

옵션

옵션품목

옵션품	옵션 형명	옵션 모델	기능(역할)
	Modbus Splitter RJ12 to RJ45	LS-LVSS-026	RJ45 cable을 RJ12 cable로 변환
	Modbus Terminating Resistor RJ45	LS-LVSS-035	Modbus 종단저항기
	Remote Keypad Kit Modbus Master	LS-LVSS-012	원격 디스플레이
	Modbus Splitter RJ12 to RJ45	LS-LVSS-011	Modbus signal 분배기
	Finger Guard kit IP20	LS-LVSS-005 (본체 : LS-LVSS-101~117) LS-LVSS-009 (본체 : LS-LVSS-201~205)	전기 충전부 접촉 방지
	Anybus M40 Ethernet/IP	LS-LVSS-6604	TCP/IP 사용
	Anybus M30 Modbus TCP	LS-LVSS-6223	TCP/IP 사용
	Anybus M30 Profibus DP	LS-LVSS-6270	TCP/IP 사용

셋팅 방법


1
Stopped
OL = 0%
I = 0.0A
P = 0.0kW
BACK
START
"Application 선택"
2
Select your Application
Default
Heavy
Agitatix
Compressor Centrifugal
Compressor Redprocating
Up
Dn
BACK
"Trip Class 확인"
3
Select your Application
Wainlnql
Selected Application will Enaide Tdp class 20
Centrifugal
Compressor Redprocating
BACK
NEXT
트립 클래스가 10보다 높은 어플리케이션을 선택한 경우에만 나타납니다.
"모터 전류 설정"
4
Set Motor Current Rating
100%
1 7 A
BACK
"Control Mode 선택"
5
Select Control Mode
Local Touch Screen
User Programmable
Two Wire Control
Three Wire Control
Modbus Network
BACK
"제어 전압 선택"
6
Select Control Voltage
230V
110V
24V
BACK
"설정값 확인"
7
Auto Setup Summary
Motor Current: 17A
Application: Compressor Screen
Control Logic: Local Touch Screen
Digital input Viltage: 24Vdc
Trip Class: 20
Save ?
BACK
No
"START"
8
Stopped
OL = 0%
I = 0.0A
P = 0.0kW
BACK
HOME

Application


Pump


Compressor


Chiller


Fan & Blower


Centrifugal Device


Conveyor

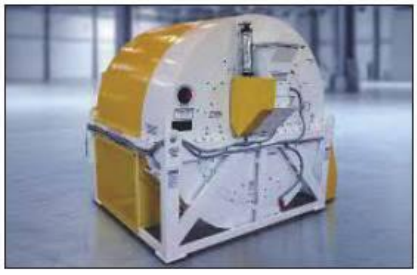

Crusher


Extruder


Wind Mill


Bow Thruster


Extractor

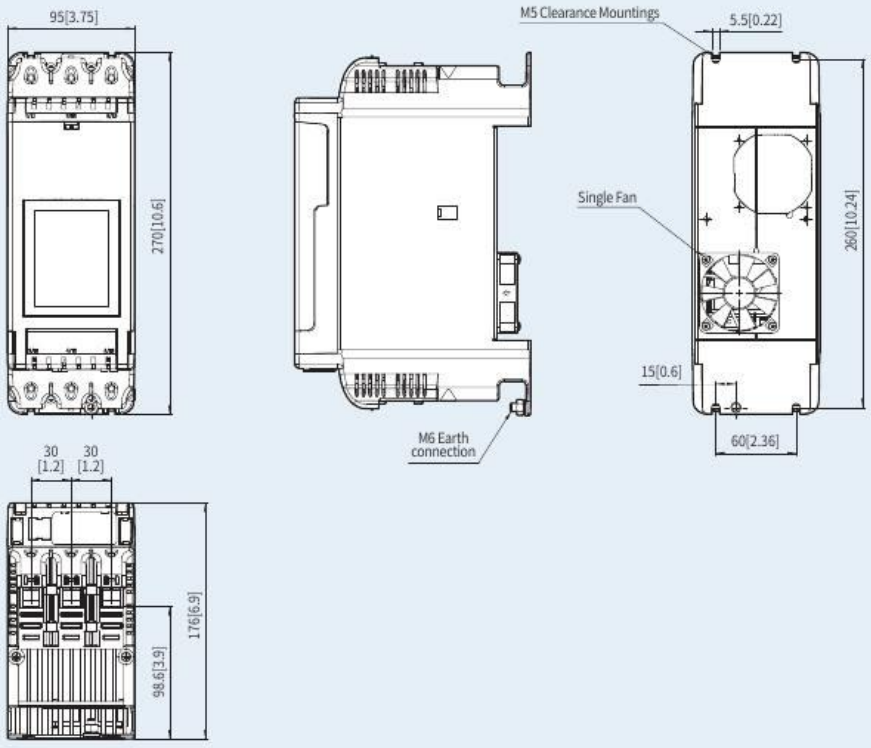

Rubber Machine


Axial Fan for Tunnel

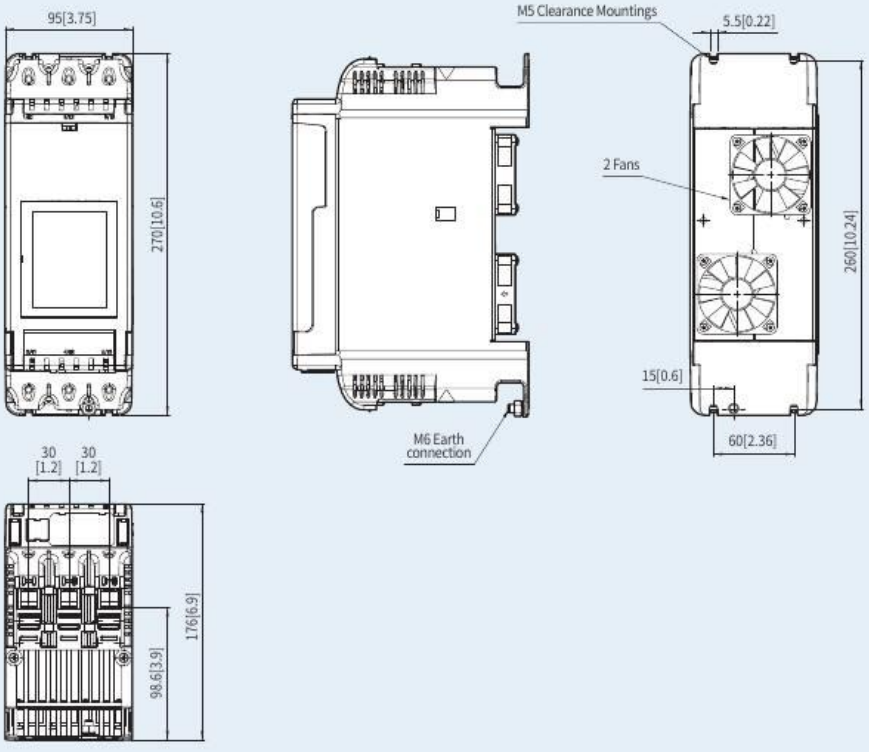
외형치수

단위 : mm[inch]

LS-LVSS-101~105



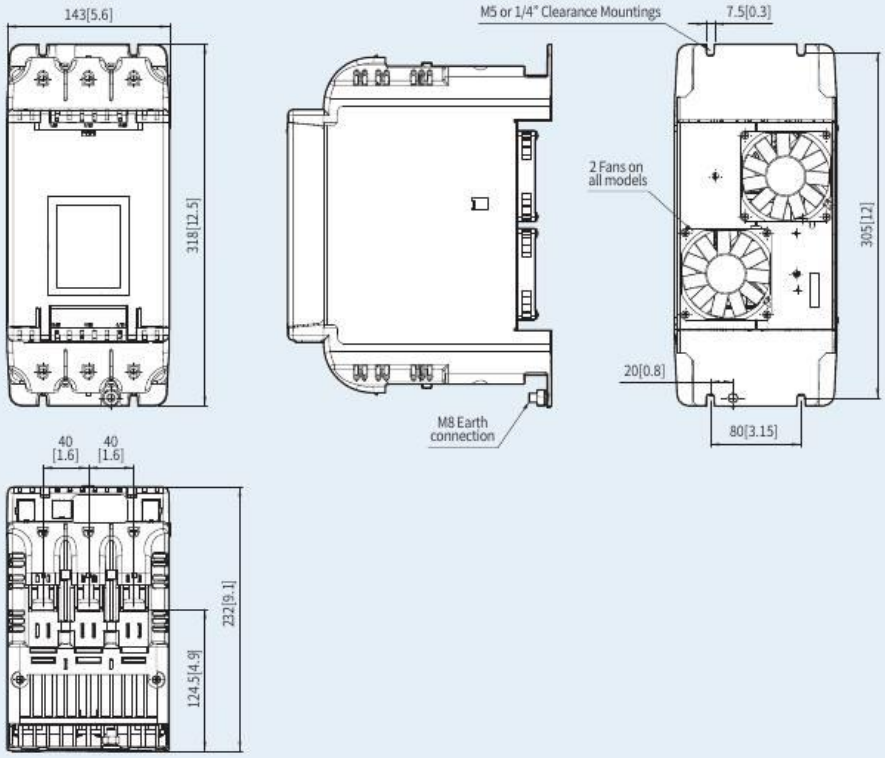
LS-LVSS-107~117



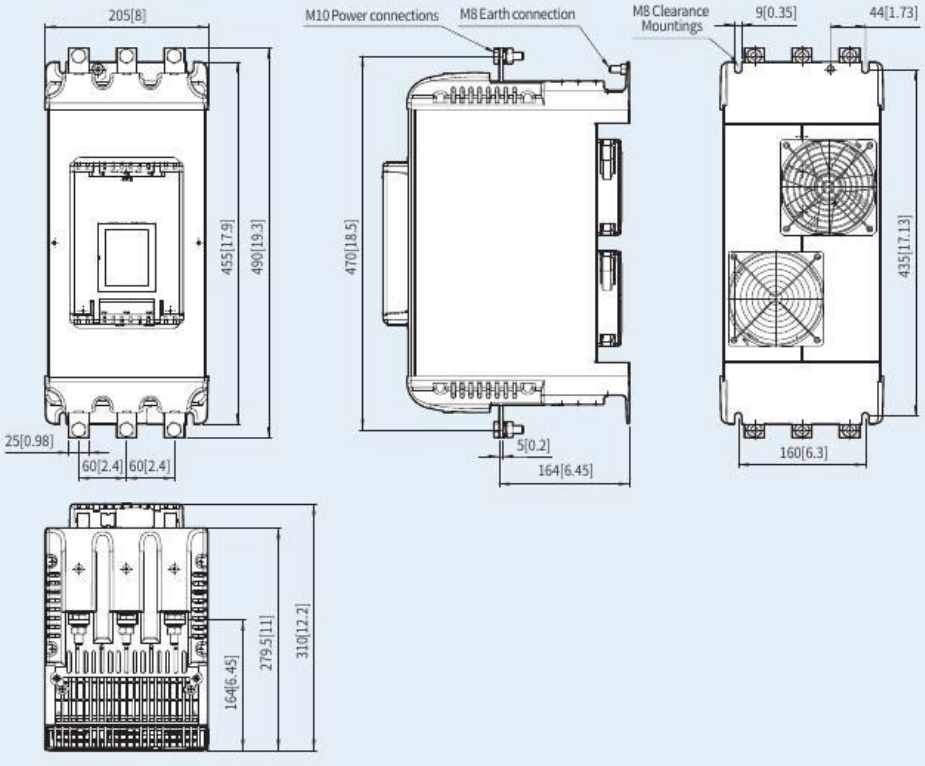
외형치수

단위 : mm[inch]

LS-LVSS-201~205

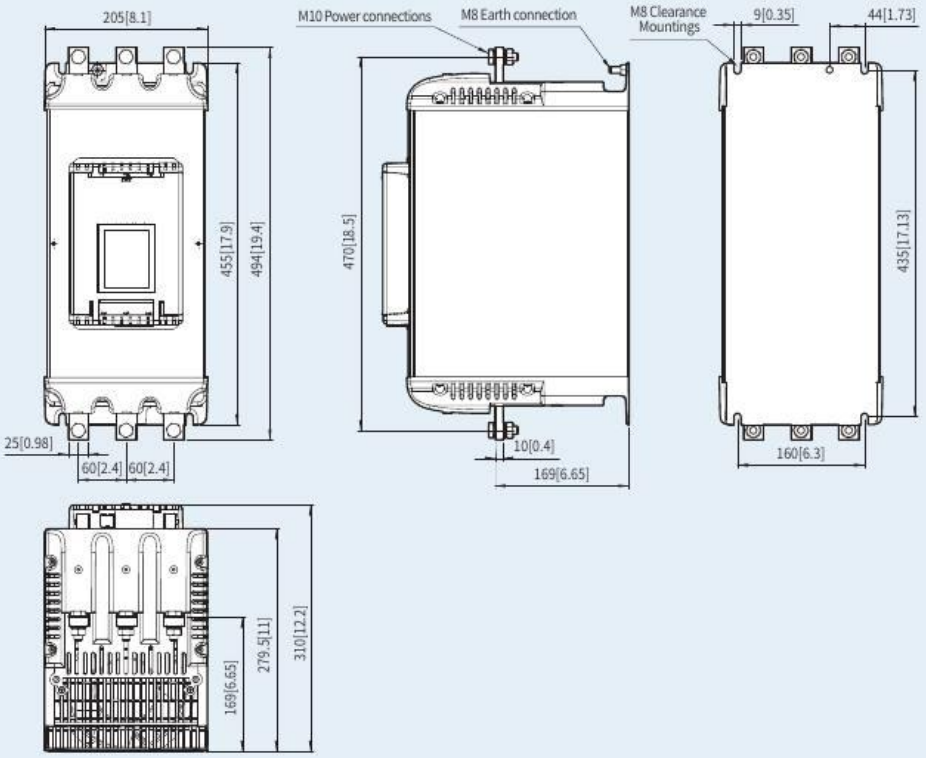


LS-LVSS-301~305

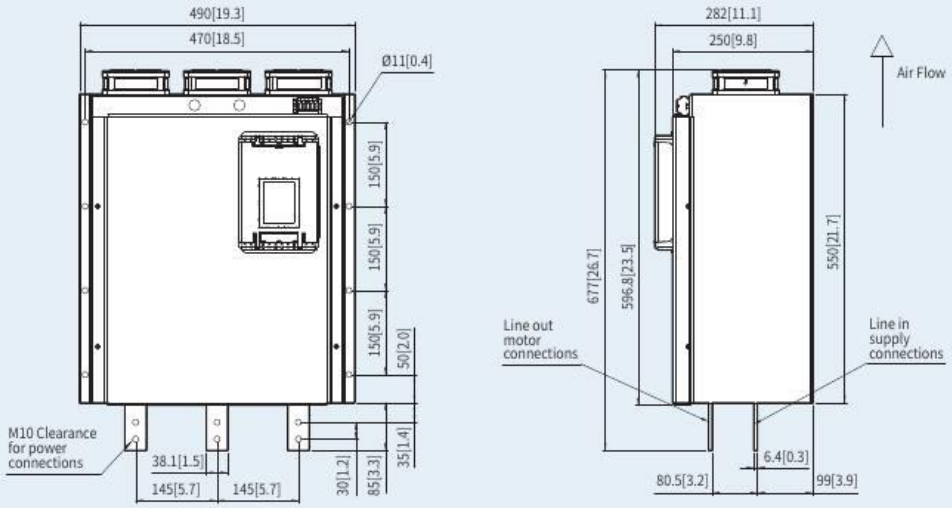


단위 : mm[inch]

LS-LVSS-307~309



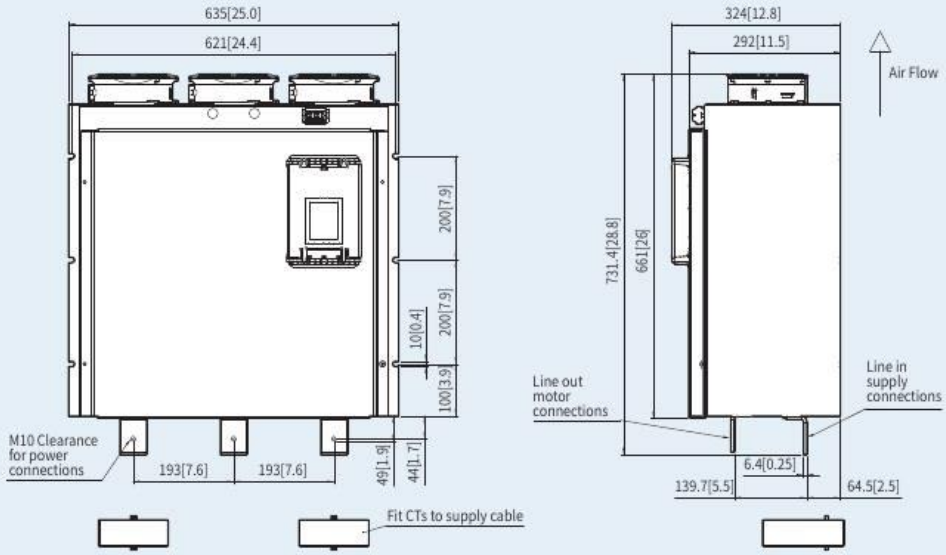
LS-LVSS-401~403



외형치수

단위 : mm[inch]

LS-LVSS-501~505



본사
경기도 안양시 동안구 엘에스로 127 LS타워

서울 사무소
서울특별시 용산구 한강대로 92 LS용산타워

구입문의

서울영업

저압/고압기기	TEL: (02)2034-4489
계전/계측기기	TEL: (02)2034-4980
계량기/원격검침	TEL: (02)2034-4581
변압기	TEL: (02)2034-4581
수배전반	TEL: (02)2034-4974
부스덕트	TEL: (02)2034-4777
초고압	TEL: (02)2034-4807
FACTS	TEL: (02)2034-4380
진단	TEL: (02)2034-4596
철도	TEL: (02)2034-4387
대전영업	TEL: (042)820-4203
	FAX: (042)820-4298
대구영업	TEL: (053)603-7711
	FAX: (053)603-7777
나주영업	TEL: (062)510-1815
	FAX: (062)526-3262
부산영업	TEL: (051)310-6811
	FAX: (051)310-6827

A/S문의

서울/경기/강원	TEL: (031)8085-2870-2
	FAX: (031)8085-2873
대전	TEL: (042)826-2080
	FAX: (042)826-2081
대구	TEL: (053)954-5241
	FAX: (053)954-5242
나주	TEL: (061)333-2080
	FAX: (061)333-2081
부산	TEL: (070)8691-1077
	FAX: (070)7589-1078

기술문의

기술상담센터	TEL: 1544-2080

교육/세미나 문의

연수원 (청주)	TEL: (043)261-6994
	FAX: (043)261-6996
세미나	TEL: (02)2034-4579
	seminar@ls-electric.com

전력기기 챗봇 서비스
언제든지 전력기기에 대한 도움이 필요할 땐 챗봇 서비스를 이용해주세요.

전력기기 제품총람
QR코드를 통해 언제 어디서든 카탈로그를 다운로드할 수 있습니다.

www.ls-electric.com

※ 본 제품의 규격은 품질개선을 위하여 예고없이 변경될 수 있으므로 제품구입시 문의 바랍니다.
ⓒ 2022. 03 LS ELECTRIC Co., Ltd. All rights reserved. / (01) 2022. 03 Staffcom